Ideals, Quotient Rings, Ring Homomorphisms, Prime and Maximal Ideals

From the class book: Chapter 15 (Sections 15.1 to 15.4), Chapter 16 (Sections 16.1 to 16.5), Chapter 17 (Sections 17.1 to 17.6)

1. Key definitions

Subring. A non-empty $S \subseteq R$ that is a ring under the same operations. Test: $a - b \in S$ and $ab \in S$ for all $a, b \in S$. (Subrings need not contain 1; $n\mathbb{Z}$ are all the subrings of $\mathbb{Z}$.)

Ideal. A subring I with $ra \in I$ and $ar \in I$ for all $r \in R,\ a \in I$ (left/right ideal: one of the two). Test: $a - b \in I$ and absorption.

Generated and principal ideals. $\langle S \rangle$ is the smallest ideal containing S. In a CRU, $\langle a_1, \ldots, a_n \rangle = \{r_1 a_1 + \cdots + r_n a_n\}$ and $\langle a \rangle = aR$. Without unity, $\langle a \rangle = \{ra + na : r \in R, n \in \mathbb{Z}\}$ (in $2\mathbb{Z}$, $\langle 4 \rangle = 4\mathbb{Z}$ but $4 \cdot 2\mathbb{Z} = 8\mathbb{Z}$).

Sum, product, radical, annihilator. $I + J = \{a + b\}$; IJ = finite sums $\sum a_k b_k$; $IJ \subseteq I \cap J$. I, J are comaximal if $I + J = R$. $\sqrt{I} = \{x : x^n \in I \text{ for some } n\}$; $\mathrm{Ann}(a) = \{x : xa = 0\}$.

Quotient ring. For an ideal I, $R/I = \{a + I\}$ with $(a + I) + (b + I) = (a + b) + I$, $(a + I)(b + I) = ab + I$. Absorption is exactly what makes the product well defined.

Maximal and prime ideals. $M \neq R$ is maximal if no ideal lies strictly between M and R. In a commutative ring, $P \neq R$ is prime if $ab \in P \Rightarrow a \in P$ or $b \in P$.

Nilradical, Jacobson radical, local ring. $N(R)$ = set of nilpotents; $J(R)$ = intersection of all maximal ideals. A CRU with exactly one maximal ideal is local (e.g. fields, $\mathbb{Z}_{p^k}$, $F[[x]]$).

Ring homomorphism. $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$; unital if $f(1) = 1$. $\mathrm{Ker}\, f = \{a : f(a) = 0\}$.

2. Main results

Theorem (ideals and units)
An ideal containing a unit is the whole ring. So a division ring has only the ideals $\{0\}$, R, and a CRU with exactly two ideals is a field. Ideals of $M_n(R)$ (R with unity) are $M_n(I)$; hence $M_n(F)$ is simple but not a division ring.

Theorem (ideals of Z)
$m\mathbb{Z} + n\mathbb{Z} = \gcd(m, n)\mathbb{Z}$, $m\mathbb{Z} \cap n\mathbb{Z} = \mathrm{lcm}(m, n)\mathbb{Z}$, $(m\mathbb{Z})(n\mathbb{Z}) = mn\mathbb{Z}$; $m\mathbb{Z} \cap n\mathbb{Z} = (m\mathbb{Z})(n\mathbb{Z})$ iff $\gcd(m, n) = 1$. $\sqrt{n\mathbb{Z}} = \mathrm{rad}(n)\mathbb{Z}$. Unions: $I \cup J$ is an ideal iff $I \subseteq J$ or $J \subseteq I$.

Theorem (correspondence)
Ideals (subrings) of R/I are exactly J/I with $J \supseteq I$ an ideal (subring) of R. So $\mathbb{Z}_n$ has $\tau(n)$ ideals (all principal, $\langle \bar{d} \rangle$, $d \mid n$), and they are also its only subrings.

Theorem (maximal and prime via quotients)
For a CRU R: M maximal $\iff$ R/M is a field; P prime $\iff$ R/P is an integral domain. Hence maximal $\Rightarrow$ prime, and $\{0\}$ is prime iff R is an integral domain. (Krull) every proper ideal of a ring with unity lies in a maximal ideal.

Proof idea: if $a \notin M$, $M + \langle a \rangle = R$, so $1 = m + ra$ and $r + M$ inverts $a + M$.

Theorem (prime implies maximal)
Every prime ideal is maximal in a finite CRU, in a Boolean ring, and in a CRU with $a^n = a$ ($n > 1$) for all a; in a PID every non-zero prime ideal is maximal. If P is prime and $IJ \subseteq P$, then $I \subseteq P$ or $J \subseteq P$.

Theorem (radicals)
In a CRU, $N(R) = \bigcap$(prime ideals) and $R/N(R)$ has no non-zero nilpotents; $a \in J(R) \iff 1 - ab$ is a unit for all b; $N(R) \subseteq J(R)$, with equality in a finite CRU. Examples: $N = J = \langle p_1 \cdots p_k \rangle$ in $\mathbb{Z}_n$; $N(F[[x]]) = 0$, $J(F[[x]]) = \langle x \rangle$; $N = J = 0$ for $\mathbb{Z}$, $F[x]$, $C[0,1]$.

Theorem (maximal ideals of C[0,1])
They are exactly $M_c = \{f : f(c) = 0\}$, $c \in [0,1]$, and $C[0,1]/M_c \cong \mathbb{R}$ via $f \mapsto f(c)$. (Compactness is essential; $C(\mathbb{R})$ has other maximal ideals.)

Theorem (homomorphisms)
$\operatorname{Ker} f$ is an ideal; f is one-one iff $\operatorname{Ker} f = \{0\}$. $f(R)$ is a subring; preimages of ideals are ideals, but images of ideals need not be ($2\mathbb{Z} \subset \mathbb{Q}$). $f(1)$ is an idempotent, equal to 1 if f is onto. A homomorphism from a field is zero or injective.

Theorem (isomorphism theorems)
(First) $R/\operatorname{Ker} f \cong f(R)$. (Second) $(S + I)/I \cong S/(S \cap I)$. (Third) for $I \subseteq J$, $(R/I)/(J/I) \cong R/J$.

Theorem (Chinese Remainder Theorem)
If $I_1, \ldots, I_n$ are pairwise comaximal ideals of a CRU, $R/(I_1 \cap \cdots \cap I_n) \cong R/I_1 \times \cdots \times R/I_n$. So $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ iff $\gcd(m,n) = 1$, and $F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle \cong \prod F[x]/\langle f_i^{e_i} \rangle$ for distinct monic irreducibles f_i.

Theorem (counting homomorphisms)
A homomorphism from $\mathbb{Z}$ is fixed by the idempotent $e = f(1)$: $\#\mathrm{Hom}(\mathbb{Z}, S)$ = number of idempotents of S. A homomorphism $\mathbb{Z}_m \to \mathbb{Z}_n$ is $\bar{x} \mapsto xe$ with $e^2 = e$, $me = 0$ in $\mathbb{Z}_n$;

$$\#\mathrm{Hom}(\mathbb{Z}_m, \mathbb{Z}_n) = 2^{\,\omega(n) - \omega(n/\gcd(m,n))}.$$

A unital one exists iff $n \mid m$, and is then unique.

Theorem (field of quotients)
Every integral domain D embeds in a smallest field $Q(D) = \{a/b : b \neq 0\}$, unique up to isomorphism. $Q(\mathbb{Z}) = \mathbb{Q}$, $Q(\mathbb{Z}[i]) = Q(\mathbb{Z}[2i]) = \mathbb{Q}(i)$, $Q(F[x]) = F(x)$, $Q(\mathbb{Z}[x]) = \mathbb{Q}(x)$; a finite domain is its own quotient field. Every field has a prime subfield $\cong \mathbb{Q}$ or $\mathbb{Z}_p$.

3. Standard tables

Counting ideals

Ring	Number of ideals	Example
field; $M_n(F)$ (two-sided)	2	$\mathbb{Q}$, $M_2(\mathbb{R})$
$\mathbb{Z}_n$	$\tau(n)$	$\mathbb{Z}_{36}$: 9
$R_1 \times R_2$ (with unity)	product (ideals are $I_1 \times I_2$)	$\mathbb{Z}_{12} \times \mathbb{Z}_8$: 24
$F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle$	$(e_1+1)\cdots(e_r+1)$	$\mathbb{Q}[x]/\langle x^3 - x \rangle$: 8
$M_n(\mathbb{Z}_m)$	$\tau(m)$	$M_2(\mathbb{Z}_6)$: 4
left ideals of $M_2(\mathbb{F}_q)$	$q+3$	$M_2(\mathbb{Z}_2)$: 5

Maximal and prime ideals; quotients

Ring	Maximal ideals	Prime, not maximal
$\mathbb{Z}$	$p\mathbb{Z}$	$\{0\}$
$\mathbb{Z}_n$	$\langle \bar{p} \rangle$, $p \mid n$ ($\omega(n)$ of them)	none
$F[x]$	$\langle f \rangle$, f irreducible	$\{0\}$
$\mathbb{R}[x]$	$\langle x-a \rangle$, $\langle x^2+bx+c \rangle$ with $b^2 < 4c$	$\{0\}$
$\mathbb{Z}[x]$	$\langle p, f \rangle$, f irreducible mod p	$\{0\}$, $\langle p \rangle$, $\langle f \rangle$ (f irreducible)
$\mathbb{Z} \times \mathbb{Z}$	$p\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times p\mathbb{Z}$	$\mathbb{Z} \times \{0\}$, $\{0\} \times \mathbb{Z}$

Quotient	Order	Field / domain iff
$F[x]/\langle f \rangle$	$\lvert F \rvert^{\deg f}$	f irreducible (both)
$\mathbb{Z}[x]/\langle p, f \rangle \cong \mathbb{Z}_p[x]/\langle \bar{f} \rangle$	$p^{\deg f}$	$\bar{f}$ irreducible mod p
$\mathbb{Z}[x]/\langle f \rangle$, $\deg f \geq 1$	infinite	never a field; domain iff f irreducible in $\mathbb{Z}[x]$
$\mathbb{Z}[i]/\langle n \rangle$	n^2	n prime, $n \equiv 3 \pmod 4$
$\mathbb{Z}[i]/\langle a+bi \rangle$	a^2+b^2	$a+bi$ a Gaussian prime

Standard isomorphisms: $\mathbb{R}[x]/\langle x^2+1 \rangle \cong \mathbb{C}$; $\mathbb{Z}[x]/\langle x \rangle \cong \mathbb{Z}$; $\mathbb{Z}[x]/\langle n, x \rangle \cong \mathbb{Z}_n$; $\mathbb{Z}[x]/\langle x^2+1 \rangle \cong \mathbb{Z}[i]$; $\mathbb{Z}[i]/\langle 2-i \rangle \cong \mathbb{Z}_5$ (via $a+bi \mapsto a+2b$); $\mathbb{Q}[x]/\langle x^2-2 \rangle \cong \mathbb{Q}(\sqrt{2})$; $\mathbb{R}[x]/\langle x^4-1 \rangle \cong \mathbb{R} \times \mathbb{R} \times \mathbb{C}$.

Other homomorphism counts

Set	Count
$\mathbf{Hom}(\mathbb{Z},\mathbb{Z})$, $\mathbf{Hom}(\mathbb{Q},\mathbb{Q})$, $\mathbf{Hom}(\mathbb{R},\mathbb{R})$	2 (zero and identity)
$\mathbf{Aut}(\mathbb{R})$, $\mathbf{Aut}(\mathbb{Q}(\sqrt[3]{2}))$	1
$\mathbf{Aut}(\mathbb{Q}(\sqrt{d}))$; continuous automorphisms of $\mathbb{C}$	2
$\mathbf{Hom}(\mathbb{Z} \times \mathbb{Z}, \mathbb{Z})$	3 (zero, two projections)
$\mathbf{Hom}(\mathbb{Z}[i], \mathbb{Z})$	1 (zero map only)
unital $\mathbf{Hom}(\mathbb{Z}[i], \mathbb{Z}_n)$	number of solutions of $t^2 \equiv -1 \pmod n$

4. Exam shortcuts and standard counter-examples

Exam shortcut. Decide by the quotient: field ⇒ maximal; integral domain but not field ⇒ prime, not maximal; zero divisors ⇒ not prime. To prove $R/I \cong S$, find an onto homomorphism $R \to S$ with kernel I.

Standard counter-examples. $\mathbb{Z} \subset \mathbb{Q}$ is a subring, not an ideal. $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subring. The diagonal $\{(a,a)\} \subset \mathbb{Z} \times \mathbb{Z}$ is a subring that is not a product and not an ideal. $\langle 2, x \rangle$ in $\mathbb{Z}[x]$ is maximal but not principal. Without unity, $4\mathbb{Z}$ is maximal in $2\mathbb{Z}$ but not prime ($2 \cdot 2 \in 4\mathbb{Z}$) and $2\mathbb{Z}/4\mathbb{Z}$ is not a field. "Is an ideal of" is not transitive. $\langle x^2+1 \rangle$ is maximal in $\mathbb{R}[x]$, not in $\mathbb{C}[x]$, and prime but not maximal in $\mathbb{Z}[x]$. $2\mathbb{Z}[x]$ is prime, not maximal, in $\mathbb{Z}[x]$.

Standard counter-examples. $f(n) = 2n$ on $\mathbb{Z}$ is additive, not multiplicative; $\det$ is multiplicative, not additive. Images can gain properties: $\mathbb{Z} \to \mathbb{Z}_6$ destroys the domain property; $2\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}_3$ has unity $\bar{4}$. $m\mathbb{Z} \cong n\mathbb{Z}$ as rings only if $m = n$.

Exam shortcut. $\mathbb{Z}_m \times \mathbb{Z}_n$ has $\omega(m) + \omega(n)$ maximal ideals; $F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle$ has r maximal ideals and is reduced iff all $e_i = 1$.

5. Solved examples

Example 1. Find the number of ideals and the number of maximal ideals of $\mathbb{Z}_{72} \times \mathbb{Z}_{10}$.

Solution. Ideals are products: $\tau(72)\tau(10) = 12 \cdot 4 = 48$. Maximal ideals have the form $M \times \mathbb{Z}_{10}$ or $\mathbb{Z}_{72} \times M$: $\omega(72) + \omega(10) = 2 + 2 = 4$.

Example 2. Show that $\mathbb{Z}_3[x]/\langle x^3 + 2x + 1 \rangle$ is a field and find its order.

Solution. Values at $0, 1, 2$: 1, $1 + 2 + 1 = 4 \equiv 1$, $8 + 4 + 1 = 13 \equiv 1$. No roots, and a cubic without roots is irreducible, so the ideal is maximal and the quotient is a field with $3^3 = 27$ elements.

Example 3. Classify $\langle 5, x^2+1 \rangle$ and $\langle 7, x^2+1 \rangle$ in $\mathbb{Z}[x]$.

Solution. $\mathbb{Z}[x]/\langle p, x^2+1 \rangle \cong \mathbb{Z}_p[x]/\langle x^2+1 \rangle$. For $p = 5$, $x^2 + 1 = (x-2)(x-3)$, so the quotient is $\mathbb{Z}_5 \times \mathbb{Z}_5$ (CRT), which has zero divisors: the ideal is not prime. For $p = 7 \equiv 3 \pmod 4$, -1 is not a square mod 7, so $x^2 + 1$ is irreducible and the quotient is a field of 49 elements: the ideal is maximal.

Example 4. Is $\langle 3 + i \rangle$ a prime ideal of $\mathbb{Z}[i]$?

Solution. $|\mathbb{Z}[i]/\langle 3+i \rangle| = 3^2 + 1^2 = 10$. A finite integral domain has prime power order, so the quotient (in fact $\cong \mathbb{Z}_{10}$) is not a domain. Indeed $3 + i = (1+i)(2-i)$ with neither factor a unit. The ideal is not prime.

Example 5. Find the number of ring homomorphisms $\mathbb{Z}_{20} \to \mathbb{Z}_{12}$ and $\mathbb{Z}_{15} \to \mathbb{Z}_{10}$, and list them.

Solution. $\mathbb{Z}_{20} \to \mathbb{Z}_{12}$: $\gcd = 4$, $12/4 = 3$, $2^{\omega(12)-\omega(3)} = 2^1 = 2$. Directly: idempotents of $\mathbb{Z}_{12}$ are $0, 1, 4, 9$, and $20e \equiv 0 \pmod{12}$ holds only for $0, 9$. Maps: $x \mapsto 0$, $x \mapsto 9x$. $\mathbb{Z}_{15} \to \mathbb{Z}_{10}$: $\gcd = 5$, $10/5 = 2$, $2^{2-1} = 2$: $e \in \{0, 6\}$ ($6^2 = 36 \equiv 6$, $15 \cdot 6 = 90 \equiv 0$). Neither has a unital map, as $12 \nmid 20$ and $10 \nmid 15$.

Example 6. Describe $\mathbb{Q}[x]/\langle x^4 - 1 \rangle$: its CRT decomposition, number of ideals, maximal ideals, and nilradical.

Solution. Over $\mathbb{Q}$, $x^4 - 1 = (x-1)(x+1)(x^2+1)$ with distinct irreducible factors. By CRT the quotient is $\mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}(i)$. Ideals: $2^3 = 8$ (monic divisors). Maximal ideals: 3. All exponents are 1, so the nilradical is 0. Over $\mathbb{R}$ the same ring is $\mathbb{R} \times \mathbb{R} \times \mathbb{C}$.

Example 7. Find $\sqrt{\langle 200 \rangle}$ in $\mathbb{Z}$ and the nilradical of $\mathbb{Z}_{200}$, with its size.

Solution. $200 = 2^3 \cdot 5^2$, $\mathbf{rad}(200) = 10$, so $\sqrt{\langle 200 \rangle} = \langle 10 \rangle$. Then $N(\mathbb{Z}_{200}) = \langle \bar{10} \rangle = J(\mathbb{Z}_{200})$, with $200/10 = 20$ elements.